A Novel Mapping of ECG and PPG to Ensure the Safety of Health Monitoring Applications

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***Abstract*—Biophysical signals, such as electrocardiogram (ECG) and photoplethysmogram (PPG) are key to the sensing of vital parameters for wellbeing. Coincidentally, both ECG and PPG provide a *different window* into the same phenomena, namely, the cardiac cycle. However, they are used separately for various health monitoring tasks, which often have safety implications. As these sensors are subject to errors and failures, there is a need to develop methods that consider both sensors so as to enhance the safety of the health monitoring application. Considering this, we present the first approach to establish the key relationships between ECG and PPG signals formally. We combine formal runtime monitoring with statistical analysis and regression analysis for our results.**

***Index Terms*—Electrocardiogram (ECG) and photoplethysmogram (PPG) signals, formal monitoring.**

I. Introduction

WEARABLE devices with sensors, such as electrocardiogram (ECG) and photoplethysmogram (PPG) are becoming the norm for many mobile health applications. These include cardiac health monitoring, continuous blood pressure, and blood glucose monitoring [1]–[4]. These applications use either the PPG sensor or the ECG sensor. As these applications are life critical, ensuring that they provide robust results at all times is essential. However, the issue of safety of wearable health monitoring applications has received scant attention. The objective of this letter is to propose a methodology that combines statistical analysis with formal methods [5], so as to ensure that mobile health applications with safety implications offer sound results at all times.

Coincidentally, both ECG and PPG provide a *different window* into the same phenomena, namely, the cardiac cycle. Also, in many wearable devices, both sensors are available, albeit that they are used separately. Since both devices operate on the cardiac cycle, if the ECG events and PPG events are correlated, they could be used simultaneously, especially to ensure that the application operates safely in the presence of faults in either sensor.

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A. Related Work

ECG and PPG have been widely used for exposing information related to human physiology in real time, enabling people to monitor their health and wellbeing [6], [7]. In several works, including [8] and [9], the authors have analyzed only the PPG for noninvasive BP monitoring. Similarly, in [10] and [11], diabetes monitoring using the ECG is developed. However, PPG sensors are susceptible to motion artefacts, and noise [12], [13] and ECG sensors are subject to failures [14].

Combining multiple physiological signals will enhance the robustness of monitoring health parameters, as envisaged in [15]. Thus, very recently, researchers have started examining sensor fusion using the ECG and PPG signals. In this regard, to analyze the BP using ECG, Wu *et al.* [16] have correlated ECG intervals with PPG intervals. However, this work lacks any scientific basis for the suggested mapping. Similarly, in [17], machine learning approaches to the estimation of ECG parameters using PPG features are developed. However, the formal basis for the developed correlation is lacking. Finally, in [18], the performance of heart rate monitors based on ECG and PPG is compared. However, the comparison is limited to the accuracy of the sensors and is not developed to determine the mapping between the timing intervals of ECG and PPG. Orthogonally, the ECG signal has been employed for monitoring some security policies of pacemakers [19]. A key idea is to noninvisibly monitor the body surface ECG signal to determine if a hacker has modified the desired timing intervals that a pacemaker maintains. The reliability of this approach could be enhanced by using multiple physiological signals, such as ECG and PPG using a suitable runtime verification monitor, which is designed for multiple signals. For this to work, we would also need a mapping between the ECG and PPG intervals, which is lacking in the literature.

Overview of the Proposed Approach: We study the simultaneous recordings of ECG and PPG of various subjects. These recordings are then used to extract features of the signals, enabling the extraction of key events and the associated intervals for both the signals. Based on the initial screening of the data streams of ECG and PPG (using spreadsheets), we propose a mapping between events and intervals of ECG and PPG. We perform statistical analysis for validating the proposed ECG to PPG mappings. For strengthening the proposed mapping, formal runtime monitoring (RV) is used as an additional approach. Finally, we employ regression analysis to analyze the lag time between ECG and PPG events.

The key contribution is that this letter, for the first time, establishes a mapping of certain ECG events and intervals

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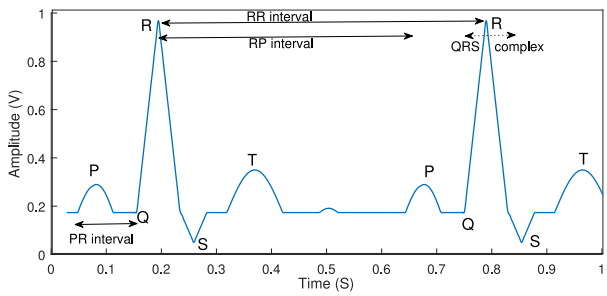

Fig. 1. Typical ECG signal.

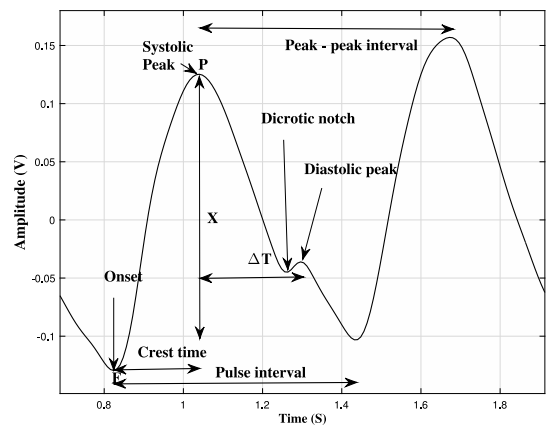

Fig. 2. Typical PPG signal.

to corresponding PPG events and intervals. While such correlation is alluded to in earlier papers [16], [17], there is no prior work that formally establishes such a mapping. Such a mapping will be extremely valuable for mobile health applications, where using sensor fusion, the safety of the application can be enhanced.

This letter is organized as follows: in Section II, we briefly present the basics of ECG and PPG, and in Section III, we propose the mapping between ECG and PPG. In Section IV, we discuss the validation of ECG to PPG mapping using formal runtime verification (RV) monitoring, statistical analysis, and regression analysis. Finally, conclusions are drawn in Section V.

II. ECG–PPG Preliminaries

A cardiac cycle represents human heart activity from one heartbeat to another. It consists of two periods: one during which the heart muscle relaxes and refills with blood, called diastole, following a period of heart contraction and pumping of blood, called systole. Both ECG and PPG present systole and diastole of a cardiac cycle.

ECG: The ECG is the waveform produced by the heart showing the electrical activity of the heart over a period of time. A typical ECG signal is illustrated in Fig. 1. The P-wave symbolizes atrial depolarization (indicating an atrial event has occurred). The PR interval is the time interval between the beginning of the P-wave and the beginning of the QRS-complex. It shows the time taken by the electrical impulses to travel between the atria and ventricles. The QRS-complex denotes the ventricular depolarization and the T-wave represents ventricular repolarization. The RR interval in ECG is the time interval between two consecutive R-peaks.

PPG: The PPG represents the variations in blood volume or blood flow in the body which goes from the heart to the fingertips and toes through the blood vessels within a cardiac cycle [20]. The various events and intervals of a PPG are shown in Fig. 2.

The time interval between the beginning and end of the PPG signal, shown in Fig. 2, is known as the *pulse interval*. The distance between two consecutive systolic peaks is referred to as the *peak-to-peak* interval. The time interval between the systolic and diastolic peaks (ΔT) is related to the time taken for the pulse wave to propagate from the heart to the periphery and back. *Crest* or *systole period* is the time taken by PPG from the onset to its systolic peak. Similarly, the time taken by the signal in the diastole phase (systolic peak to subsequent onset) is known as the *diastole period*.

III. Mapping of ECG–PPG Events and Intervals

This section presents the mappings of ECG–PPG events and intervals based on the initial manual analysis (using spreadsheets) of ECG–PPG data streams.

We studied the simultaneous recordings of ECG–PPG available in the BIDMC database in Physionet [21]. The database contains 53 recordings, each of 8 min in duration. The signals were preprocessed for the removal of noise. The signals are processed using the Neurokit2 tool[1] [22] to extract events, such as P-peak, R-peak, Q-peak, and T-peak of ECG and onset and the systolic peak of PPG. We thereby computed ECG intervals, such as PR (P-peak to R-peak), QR (Q-peak to R-peak), RR, RP (R-peak to P-peak), RT (R-peak to T-peak), QT (Q-peak to T-peak), and systole and diastole periods of PPG, considering approximately 7586 samples of the dataset.

It has been studied that the systolic peak of a PPG signal corresponds to the R-peak of an ECG signal as both reflect ventricular contraction [6]. Considering this as the starting point of our analysis, we map other events and intervals of ECG–PPG. It is observed that PR intervals of ECG are in a similar range to systole periods of PPG. Since the R-peak of ECG corresponds to the systolic peak of PPG, the P-wave of ECG may be mapped to the onset of PPG. It is also observed that the RR intervals of ECG are similar to the peak-to-peak intervals of PPG. Similarly, the RP intervals of ECG are found to be in a similar range to the diastole periods of PPG. During the analysis, it is found that there is a delay of 600–700 ms between the mapped events, such as the R-peak of ECG and the systolic peak of PPG, and the P-peak of ECG and the onset of PPG. The summary of observations from the manual analysis of ECG–PPG is as follows.

1) *Obs_1:* The RR interval of ECG may be mapped to the peak-to-peak interval of PPG.
2) *Obs_2:* The P-peak of ECG may correspond to the onset of PPG.
3) *Obs_3:* The PR interval of ECG may be mapped to the systole period of PPG.
4) *Obs_4:* The RP interval of ECG may be mapped to the diastole period of PPG.
5) *Obs_5:* The onset of PPG may lag the P-peak of ECG by 600–700 ms.
6) *Obs_6:* The systolic peak of PPG may lag the R-peak of ECG by 600–700 ms.

[1]Neurokit is a Python toolbox for physiological signal processing. It implements signal processing algorithms, such as Pan–Tomkin's algorithm, discrete wavelength transform (DWT), etc., in python.

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TABLE I
STATISTICAL ANALYSIS OF ECG AND PPG INTERVALS

ECG intervals	Systole period (onset-systolic peak interval)		Diastole period (systolic peak-onset interval)	
	Coef. r	p value	Coef. r	p value
PR	0.979	< 0.0001	-0.080	<0.0001
QR	-0.120	< 0.0001	0.016	0.153
RP	-0.023	< 0.045	0.788	< 0.0001
RT	-0.091	< 0.0001	0.216	< 0.0001
QT	-0.151	< 0.0001	0.239	< 0.0001

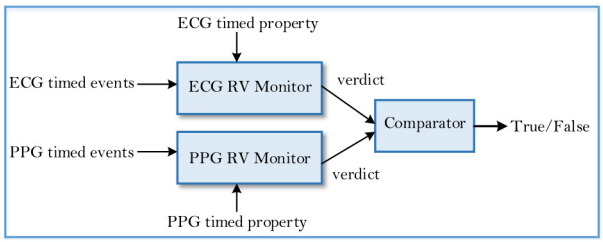

Fig. 3. RV framework for mapping ECG–PPG intervals.

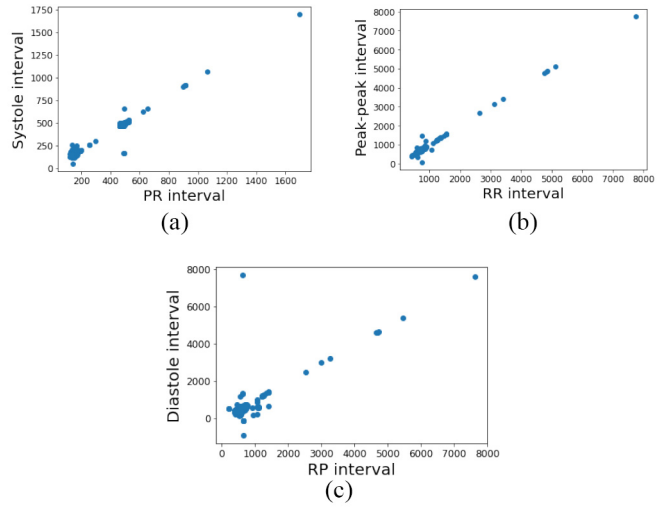

Fig. 4. Scatter plots of correlated ECG–PPG intervals. (a) PPG systole period versus ECG PR interval. (b) PPG peak-to-peak interval versus ECG RR interval. (c) RP interval of ECG versus diastole period of PPG.

IV. VALIDATION OF THE PROPOSED ECG–PPG MAPPINGS

In this section, the proposed mappings (in Section III) are validated using a runtime monitoring framework, statistical analysis, and regression analysis.

Statistical Approach for ECG–PPG Correlation: We performed Pearson's correlation analysis to evaluate the mappings between ECG intervals (RR, PR, and RP) and PPG intervals (peak-to-peak interval, systole, and diastole periods) for approximately 7586 samples of the BIDMC dataset. We plotted the scatter plots for ECG–PPG intervals shown in Fig. 4, and it exhibits a linear relationship. Also, it is found that the ECG–PPG intervals data follow the normal distribution. So, we calculate Pearson's correlation coefficient (r). The significance value (p value) is computed and $p < 0.05$ is considered to be significant. The correlational analysis is performed using the IBM SPSS version 22 framework[2] (SPSS Inc., Chicago, IL, USA) [23]. A high correlation coefficient (r) value (maximum value is 1) with a significance value of $p < 0.05$ suggests a high correlation.

The Pearson correlation coefficient r and significance value p for the RR interval of ECG and the peak-to-peak interval of PPG are found to be 0.9941 and < 0.00001, respectively, suggesting a high degree of correlation between the two (Obs_1). Pearson's correlation coefficient r and significance values p for other ECG–PPG intervals are presented in Table I. From Table I, it is evident that the PR interval of the ECG is highly correlated to the systole period of the PPG (Obs_3). This can be attributed to the fact that both the intervals represent the systole period of the cardiac cycle. Similarly, the RP interval of the ECG is correlated to the diastole period of the PPG (Obs_4). This can be explained in the light that both the intervals count toward the diastole period of the cardiac cycle.

Formal Runtime Monitoring (RV) Approach for ECG–PPG Correlation: Formal RV [24], [25], a lightweight verification approach dealing with automatic synthesis of RV, monitors from a formal high-level specification of the given set of policies. RV monitors are used to verify a stored execution of a system (offline verification) or the current live execution of a system (online) with respect to a desired correctness policy φ.

As shown in Fig. 3, we synthesis two monitors: 1) ECG and 2) PPG monitors, from the timed property of ECG (e.g., "the PR interval of ECG should be within 210 ms in a cycle") and PPG (e.g., "the systole period of PPG should be within 210 ms in a cycle"), formalized as timed automata (TA), following the approaches discussed in [24], [25]. As shown in Fig. 3, the parallel monitors are fed with events (to be mapped) along with their time of occurrence of the respective signals belonging to the same cardiac cycle for validating the mapping. The RV monitor emits verdict *true* if the property is satisfied; otherwise, *false*. At each step, the verdicts of both the parallel monitors are composed to emit the final verdict. The final verdict would be *true* when the verdicts in both the monitors are the same, *false* otherwise. The following example illustrates the behavior of the proposed RV framework for the correlations.

Example 1 (Example Illustrating Behavior of the RV Framework for ECG–PPG Interval Mapping): Let us consider a scenario of establishing the mapping of the PR interval of ECG to the onset-systolic peak interval (systole period) of PPG. This suggests monitoring the property: "the PR interval of ECG and systole period of PPG should be within 210 ms in a cardiac cycle" (considering the average PR interval of ECG as 210 ms). Consider an ECG trace (events along with the time of occurrence): [(p, 728), (r, 888)] where p and r stand for P-peak and R-peak of ECG, respectively. The PPG trace corresponding to the ECG cycle is: [(F, 1416), (P, 1568)] where F and P stand for onset and the systolic peak of PPG, respectively. It may be observed that the events in PPG are delayed, which is due to the lag between ECG and PPG events (discussed in Section III). We fed both the traces to our monitoring frameworks in parallel, i.e., ECG traces are input to the ECG monitor (resp. PPG). We maintained a guard of 210 ms in both the monitoring clocks. This is repeated in a loop throughout the recording. The monitor emits *false* whenever there is a violation. The fact that both the monitors agree in more than 90% cases establishes the proposed mapping. We adopted a similar approach for mapping other intervals.

Regression Analysis of ECG–PPG Lag Time: We performed linear regression analysis to study the mapping (Obs_2) and lag

[2]IBM SPSS Statistics is a software package used for interactive, or batched, statistical analysis [23].

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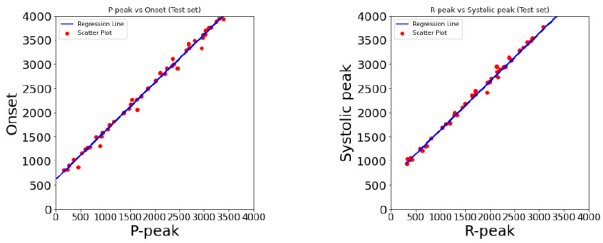

(a) (b)

Fig. 5. Regression analysis of correlated ECG–PPG events. (a) P-peaks of ECG versus onsets of PPG. (b) R-peaks of ECG versus systolic peaks of PPG.

time between ECG and PPG events (Obs_5, Obs_6). We perform regression analysis between the ECG–PPG events (to be mapped), such as P-peaks of ECG and onsets of PPG; similarly, R-peaks of ECG and systolic peaks of PPG. We implemented a linear regression model based on the regression equation, $T_{\text{ppg}} = b_1 \times T_{\text{ecg}} + lag_time$, where: T_{ecg} and T_{ppg} represent the timestamps of a correlated ECG event (e.g., R-peak) and PPG event (e.g., systolic peak), respectively, b_1 denotes the slope of the regression line, and *lag_time* denotes the lag time between the ECG–PPG event under analysis. The model is implemented in Python 3.8 using the package scikit-learn.

Fig. 5(a) represents the linear regression between the P-peaks of ECG and the onsets of PPG. The estimated coefficients are found to be 0.999 (b_1) and 607.065 ms (*lag_time*) for the regression analysis. The value of the coefficient $b_1 = 0.999$ suggests the mapping between the two events (Obs_2), and the value of the coefficient $lag_time = 607.065$ ms suggests the lag time to be in the range of 600–700 ms, validating the analysis (Obs_5). Similarly, Fig. 5(b) represents the linear regression between the R-peaks of ECG and the systolic peaks of PPG. The coefficients are found to be 0.999 (b_1) and 606.486 ms (*lag_time*) for the regression analysis. The value of the coefficient $b_1 = 0.999$ suggests the mapping between the two events, and the value of the coefficient $lag_time = 606.486$ ms suggests the lag time to be in the range of 600–700 ms, validating the observation (Obs_6).

So, using the above analysis, it is concluded that observations, Obs_1: The RR interval of ECG is mapped to the peak-to-peak interval of PPG, Obs_2: The P-peak of ECG corresponds to the onset of PPG signal, Obs_3: The PR interval of ECG is mapped to the systole period of PPG, Obs_4: The RP interval of ECG is mapped to the diastole period of PPG, Obs_5: The onset of PPG lags the P-peak of ECG by 600–700 ms, and Obs_6: The systolic peak of PPG lags the R-peak of ECG by 600–700 ms, are valid and, hence, represent the mappings between ECG and PPG events and intervals.

V. Conclusion

Two key senors, namely, the ECG and PPG are increasingly used in many mobile health applications. For enhanced safety of such applications, it is essential to use sensor fusion. To facilitate such a sensor fusion, we proposed the first novel mapping between ECG and PPG events and intervals. We validated the proposed mapping using formal monitoring and statistical approaches. Such mappings will provide a better insight into understanding the health parameters of an individual. In the future, we will develop several health monitoring applications which utilize this mapping.

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